

JAYAMALA B

Full Stack Developer | Python · MERN Stack · PHP · REST APIs · Sarvam AI · JWT
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PROFESSIONAL SUMMARY

Full Stack Developer with nearly 2 years of production experience building backend systems, REST APIs, and database-driven web applications. Proficient in Python, PHP (Laravel, CodeIgniter), and the MERN stack (MongoDB, Express.js, React.js, Node.js) with JWT-based authentication. Built a voice-enabled multilingual chatbot using Sarvam AI (Saras STT and Bulbul TTS) supporting English and Tanglish — deployed in a production customer-facing interface. Familiar with RAG (Retrieval-Augmented Generation) concepts and actively exploring LangChain. Experienced in Agile/Scrum teams using Git and Jira. Strong problem-solver with a B.Tech in AI and Data Science (CGPA 8.8) — eager to take ownership and build real-world products.

TECHNICAL SKILLS

Languages: Python (proficient), JavaScript (ES6), PHP, SQL

MERN Stack: MongoDB, Express.js, React.js, Node.js, JWT authentication

PHP Frameworks: Laravel, CodeIgniter, Core PHP

Backend & APIs: REST API design and integration, Flask, ServiceNow, Postman (API testing), JWT

Databases: MySQL (production — schema design, query optimisation, data integrity), MongoDB (NoSQL — MERN projects)

AI & NLP: Sarvam AI (Saras STT, Bulbul TTS), multilingual NLP (English, Tanglish), chatbot development, RAG concepts

Data & Reporting: Power BI dashboards, ETL, data transformation

AI Dev Tools: Claude (Anthropic), GitHub Copilot — daily usage for code generation, debugging, and documentation

Tools & Practices: Git/GitHub/GitLab, Jira, Agile/Scrum, technical documentation, debugging, code reviews

Exploring: LangChain, RAG pipelines, Docker, TypeScript

WORK EXPERIENCE

Software Developer & Data Analyst | Neuralarc Global Pvt. Ltd., Coimbatore

Jul 2024 – Present

- Built and maintained REST API integrations in Python connecting multiple backend services — ensuring reliable data flow, structured error handling, and data integrity across production workflows.
- Developed a voice-enabled multilingual chatbot using Sarvam AI's Saras (STT) and Bulbul (TTS) models — supporting English and Tanglish speech interaction, deployed in a production customer-facing interface.
- Applied RAG (Retrieval-Augmented Generation) concepts to enhance chatbot response quality with context-aware, document-grounded answers — actively deepening practical LangChain expertise.
- Built backend automation scripts using ServiceNow integrations to streamline operational processes — significantly reducing manual effort and improving production system reliability.
- Managed MySQL databases in production — designed schemas, wrote optimised queries, resolved data integrity issues, and enforced access controls across live systems.
- Delivered Power BI dashboards providing operational metrics to product and business teams; maintained comprehensive technical documentation for all systems and APIs.

Business Analyst Intern | Digital Garage, Coimbatore

Feb 2024 – May 2024

- Collaborated within Agile/Scrum delivery cycles at a product startup using Jira — contributed to sprint planning and cross-functional delivery with design and engineering teams.
- Translated product requirements into clear BRDs and user stories, bridging communication between engineering and non-technical stakeholders.

Data Science Intern | Cube N Square, Chennai

Aug 2023 – Jan 2024

- Wrote Python scripts for data preprocessing and analysis; to build and evaluate ML models — reinforced clean coding and debugging practices.
- Produced technical documentation on data science workflows for varied audiences, demonstrating ability to communicate complex topics clearly.

KEY PROJECTS

Multilingual Voice Chatbot — Python, Sarvam AI (Saras STT + Bulbul TTS), RAG concepts

- Built a production voice-enabled chatbot supporting English and Tanglish — integrated Sarvam AI's Saras STT for speech recognition and Bulbul TTS for natural voice responses, deployed in a customer-facing interface.
- Applied RAG concepts to ground chatbot responses in internal documents, improving contextual accuracy and relevance of answers for real user queries.
- structured validation of auth flows and CRUD operations.

Inventory Management System (MERN Stack) — *MongoDB, Express.js, React.js, Node.js, JWT, Postman*

- Built a complete inventory management system on the MERN stack — JWT-secured REST APIs, MongoDB document schema design, and React.js frontend for real-time stock tracking.
- Applied secure API practices throughout — JWT token validation and error handling across all endpoints tested via Postman.

ERP Module Development — *PHP (CodeIgniter), MySQL, REST APIs*

- Developed and maintained backend ERP modules with REST API integrations in CodeIgniter — built frontend views, optimised MySQL queries, and debugged cross-layer integration issues across API, backend, and database.

EDUCATION

B.Tech – Artificial Intelligence and Data Science

2020 – 2024

Arjun College of Technology, Coimbatore | CGPA: 8.8

CERTIFICATIONS & ACHIEVEMENTS

NPTEL ELITE with Silver – Data Analytics with Python (80%) | NPTEL ELITE – Data Science for Engineers | Data Science with Python – Great Learning | Gold Award – QCFI Coimbatore & Chennai Chapter | TCS NQT Cognitive Exam – 62%